A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 08

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Single genes can be transferred from any species, not just closely related ones that produce fertile hybrids **1** Desired characteristics introduced without linked unwanted ones – faster **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use	2
01.2	**1 mark each** 1 mark for each of the 3 points below. **1 Bt:** reduces insecticide spraying so fewer non-target deaths, lower cost / energy **1 Roundup Ready:** weeds controlled by spraying glyphosate without harming the crop, so higher yields with less cultivation / tillage **1 Golden rice:** provides vitamin A in the staple diet, reducing about 500 000 cases of blindness a year in India / sub-Saharan Africa **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 4 points below. **1** Gene transfer by pollen to conventional / organic crops or wild relatives (eg Bt gene killing insects feeding on wild plants) **1** Antibiotic-resistance marker genes could transfer to pathogenic bacteria – possible allergies – gene transfer to gut bacteria **1** Detection in the human food chain from crops approved only for animal feed **1 Economic:** seeds must be bought each year from patent-holding companies, raising costs and reducing local crop diversity and seed saving in LEDCs – herbicide-resistant crops may increase herbicide use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use	4
01.4	1 mark for the point below. **1** Public / political concern that risks are not fully assessed - more evidence needed (precautionary principle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 323-324; AQA spec 3.5.1.2.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Environmental DNA (eDNA) analysis	1
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Transmitters are too heavy for small birds / their batteries would not last for a long-term study **1** Geolocator tags are small and only record time and light levels – latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each separate (regional) population has its own unique gene pool **1** DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
03.1	**1 mark each** 1 mark for each of the 4 points below. **1 General physiological solvent:** most biological reactions take place between substances dissolved in water **1** Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes **1** High specific heat capacity moderates temperature changes – evaporation absorbs heat (cooling) **1 Anomalous expansion:** ice floats so water bodies do not freeze solid – absorbs UV protecting early marine life – provides aquatic habitats **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Anomalous** → an outlier / a result that does not fit the pattern **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty	4
03.2	**1 mark each** 1 mark for each of the 4 points below. **1** From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen – at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) **1** Oxygen in the stratosphere absorbed UV, forming ozone (O3) in a dynamic equilibrium of O, O2 and O3 **1** The ozone layer reduced UV reaching the surface so organisms could survive out of the water **1** Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved – many anaerobic archaea died out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	4

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Photosynthesis removed CO2 (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels **1** This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Irrigation depletes rivers and aquifers (eg Aral Sea, Ogallala) – increases evaporation **1** Soil erosion and compaction reduce water retention and infiltration so runoff and river flow fluctuations increase **1 Evapotranspiration changes:** increased in irrigated arid areas, decreased where forests were cleared **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface • **Evapotranspiration** → evaporation plus transpiration **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	3
04.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Seasonal / local:** avoids energy for heating, lighting greenhouses out of season and long-distance transport (food miles) and the pollution this causes **2 Organic:** uses natural processes (rotation, legumes, manure, biological control, predator habitats) instead of artificial fertilisers and pesticides, so less pollution, energy use and harm to non-target species – free-range and Fairtrade also accepted for welfare / social reasons **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** Food shortages after 1945 (dependence on US food aid) – farmers could not afford loans to invest so grants funded machinery, drainage, hedgerow removal, liming, improved livestock **1 Guaranteed market / price support:** government bought surpluses to hold prices up (and sold stores in poor years) giving farmers security to increase output **1 Consequences:** loss of hedgerows and wetlands, intensification, pesticide and fertiliser pollution – later food surpluses (butter mountains) that could not be sold without harming LEDC producers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-333; AQA spec 3.5.1.3-3.5.1.4.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Hydrosere **1** Single-celled algae (and floating plants) - arrive on birds / insects / from soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pioneer species** → the first species to colonise bare ground	2
05.2	**1 mark each** 1 mark for each of the 5 points below. **1** Pioneers and later plants grow and die – dead organic matter plus sediment washed in fill the pond so water becomes shallower **1** As depth falls rooted plants (pondweeds then emergent reeds with leaves above water) can anchor and colonise – emergents shade submerged plants which die out **1** Sediment / soil accumulates and becomes able to support larger plants – water-tolerant trees (willow, alder) colonise the waterlogged soil **1** Transpiration by trees and further sediment dry the soil and make it aerobic so oak can colonise – the denser canopy shades out smaller plants **1** Each community changes abiotic conditions (depth, light, nutrients, oxygen), making it suitable for the next colonisers which out-compete them – conditions become less extreme and biotic factors increasingly important **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogged** → soil pores filled with water so there is no air • **Transpiration** → water lost as vapour from the leaves • **abiotic** → non-living / physical factors	5

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Starting conditions (bare rock vs open water) differ but each stage makes conditions less extreme and builds soil / organic matter **1** The final community is controlled by the regional climate (temperature, water availability), which is the same for both - climatic climax (eg oak woodland in the UK) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **climax community** → the stable final community **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2
05.4	1 mark for the point below. **1** Lichens and algae colonise bare rock (extreme temperatures, no soil, little water) – rock fragments and dead organic matter accumulate – mosses, then grasses and ferns as thin soil forms – wind-dispersed trees (birch) then animal-dispersed (oak, beech) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 73-76; AQA spec 3.1.3.3.	
06.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Sinkers / weighted lines so hooks sink quickly below the reach of albatrosses **1** Setting lines at night when birds are not feeding **1** Bird-scaring streamers / decoys **1** Curved hooks that catch tuna but not birds **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 3 points below. **1 Effort:** limits on boat size, engine power, days at sea, number of licences – decommissioning payments to reduce fleet size where overfishing is unavoidable **1 Methods:** bans or length limits on drift nets – bans on demersal trawling over sensitive seabed (Rockall Bank, Hatton Bank deep-water reefs) **1** Shorter trawl tows so turtles / by-catch survive (over 99% survive under 10 minutes) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **decommissioning** → safely dismantling a power station at the end of its life **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	3
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Eggs / larvae are reared in captivity through the planktonic stage when most die (predation, failure to find habitat) **1** Juveniles are released when survival chances are good, increasing recruitment to the adult population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Egg / larval survival varies greatly between years so adult numbers do not predict recruitment **1** Several poor years would lower the MSY – egg surveys allow catches to be reduced in advance rather than after a sudden collapse **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **MSY** → the largest catch that can be taken without reducing the stock	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 4 points below. **1** Area must be large enough to support viable populations (breeding numbers high enough, gene pool large enough to avoid inbreeding) – species higher up food chains need larger areas (lower density) **1** Large areas contain a full range of resources (eg different fruiting trees / feeding areas through the year) **1 Shape:** the perimeter has edge conditions (a mix of neighbouring habitats) – a compact shape maximises core habitat for species needing core conditions **1** Some species benefit from edges (higher light) or small isolated habitats (ponds too small for predatory fish suit amphibians) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	4
07.2	**1 mark each** 1 mark for each of the 3 points below. **1 Varied topography:** banks, lagoons, islands, ditches created with machinery give a range of water depths (emergent, floating-leaved, submerged plant zones) **1** Islands provide roosting / nesting sites safe from terrestrial predators (foxes) **1** Control of salinity (eg intertidal conditions for opossum shrimps feeding dunlin and avocet) – scrapes for skylarks – different water depths for waders and ducks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Mobile species (birds, flying insects) colonise easily **1** Less mobile species (many plants, amphibians, invertebrates) may not reach an isolated site so must be introduced – keystone species especially **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **keystone species** → a species whose loss changes the whole community	2
07.4	1 mark for the point below. **1** Allows dispersal of surplus young, mixing of gene pools (less inbreeding) and recolonisation if a population dies out – avoids hazards (eg rope bridges for howler monkeys over roads) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-42; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Σn(n - 1) = (40 x 39) + (30 x 29) + (20 x 19) + (10 x 9) = 1560 + 870 + 380 + 90 = 2900 **1** N(N - 1) = 100 x 99 = 9900 **1** D = 9900 / 2900 = 3.41 (accept 3.4) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Simpson** → a diversity index that uses both the number of species and their abundance **Do not credit** • A bare wrong answer with no working. • 'more species always means a higher index' - the evenness of the abundances matters as well	3
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B **1** The index does not identify which species are present / a rare or important species may be missed – depends on identification skill and sampling method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 4 points below. **1** Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins) – wastes of one process are raw materials for another so nothing is depleted or accumulates **1** Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials **1 Human systems are linear:** extract - process - manufacture - use - discard – minerals are dispersed after use or mixed so recovery is difficult **1** Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up – fossil fuel use is inherently linear and unsustainable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life **1** They leach from landfill or are released by incineration – persistent, bioaccumulating **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **incineration** → burning waste, usually with energy recovery • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
09.3	**1 mark each** 1 mark for each of the 4 points below. **1** Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) **1** Easy identification of materials (coding) and separation of components so they can be recycled **1** Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) **1** Use of recycled and recyclable / biodegradable materials – reduced packaging and material use (Cradle to Cradle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers	4

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	
10.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Zooxanthellae are photosynthetic algae living inside the polyp cells **1** The algae absorb CO_2 and excretory wastes from the polyp for photosynthesis and growth (efficient nutrient recycling) **1** They supply the polyp with oxygen and glucose - over 95% of its food energy **1** In return the algae gain a protected habitat and nutrients they could not absorb from the nutrient-deficient water **1** Polyps also feed on plankton captured by cilia and nematocysts **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	4
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** Corals deposit a skeleton of calcium carbonate (calcite) **1** Each carbonate molecule requires the absorption of a molecule of carbon dioxide, so reefs are a carbon sink **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
10.3	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Importance:** cover 1% of ocean area but support over a quarter of marine species – fisheries (fish, lobsters, conch) – medicines (AZT, Ara-C, dolastatin, secosteroids, bryostatin, trabectedin, bone grafts) – climate control as carbon sink – erosion protection of low-lying islands – tourism income • **Threats:** physical damage (anchors, divers, litter, coral extraction), collection of ornaments (CITES Appendix II), sedimentation from rivers and coastal development, pollution (oil, nutrients causing algal shading, sewage, pesticides, sunscreen), ocean acidification, bleaching from warming, overfishing (giant triton / crown of thorns starfish), introduced species (red lionfish), loss of mangroves and seagrass nurseries • **Conservation:** marine protected areas / national parks (Great Barrier Reef, Belize, Chagos), mooring buoys, diver education, No Take Zones which can increase catches elsewhere, sustainable ecotourism and fishing, protection of mangroves and seagrass, erosion and pollution control, planning control of ports and resorts • **Fishing controls:** closed seasons, minimum and maximum catchable sizes, bans on dynamite, trawling and small-mesh nets • **Evaluation:** local measures cannot control global threats (warming, acidification) which need international action – enforcement in remote areas is difficult – conflicts with local livelihoods – large vs small NTZs • Justified conclusion **Also accept**	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **CITES** → the treaty that controls international trade in endangered species • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	
	Model answer - Level 3 (9 marks)	
	Tropical coral reefs cover about one per cent of the oceans yet support over a quarter of marine species, a diversity built on the symbiosis between polyps and zooxanthellae that recycles nutrients in nutrient-poor water. They matter to people for fisheries of fish, lobsters and conch; for medicines such as AZT, Ara-C, dolastatin, secosteroids, bryostatin and trabectedin discovered in reef organisms; as a carbon sink, since every carbonate unit in coral skeleton absorbs a molecule of carbon dioxide; for dissipating wave energy that would erode low-lying islands; and for tourism income.	
	They are also unusually vulnerable because they need narrow abiotic conditions. Anchors, divers, litter and coral extraction kill polyps; ornaments are collected; sediment from rivers and coastal development smothers them; oil, sewage and nutrients cause algal shading; ocean acidification hinders skeleton formation; warming causes bleaching; overfishing of giant tritons lets crown-of-thorns starfish destroy reefs; introduced lionfish eat native fish; and loss of mangroves and seagrass removes the nurseries and sediment traps reefs depend on.	
	Marine protected areas such as the Great Barrier Reef, Belize and Chagos combine mooring buoys, diver education, no-take zones that raise catches outside them, sustainable tourism and protection of linked habitats, while fishing controls set closed seasons, minimum and maximum sizes and ban dynamite, trawling and small mesh. These work well against local threats where they are enforced, and no-take zones have measurable benefits, but they cannot touch warming and acidification, which need international action, enforcement in remote areas is weak, and restrictions conflict with local livelihoods. Local conservation buys time; only global climate action can secure the reefs.	
	Weaker answer - Level 1 (3 marks)	
	Coral reefs are important because lots of fish live there and people catch them for food. Tourists visit reefs to dive and this brings money. Reefs also protect the coast from waves. Reefs are threatened by climate change which causes bleaching, and by pollution and damage from boats. They are conserved by making marine protected areas where fishing is banned. This helps but climate change is still a problem.	
	What makes the difference	
	• Level 1 has the right headings but Level 3 explains each (why reefs are diverse, why they are a carbon sink, how each threat acts).	
	• Level 3 uses named reserves and specific management measures; Level 1 gives one generic measure.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3's evaluation separates local from global threats and reaches a nuanced judgement; Level 1 states that climate change is 'still a problem'.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	
11.1	**Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Sustainability definitions – population and per capita impact – attitudes of rural and urban societies • **Negative feedback:** cloud cover and albedo, CO_2 and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • **Tipping points:** rare under negative feedback – large events (Permian methane hydrate) – self-sustaining change from human activities • **Diversity and resistance:** reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides) – links to climate change chapter – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Need:** population possibly 12 billion – humans raise carrying capacity – trophic level of diet • **Impacts to be reduced:** habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • **Social, economic and political influences:** consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • **Strategies:** cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • **Limits:** little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25

Do not write outside the box